

Avi Vyas

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🔗 github.com/avi-vyas 🔗 avi-vyas.github.io/portfolio-application

PROFESSIONAL EXPERIENCE

Software Engineer (Backend Engineer)

01/2025 – Present | Noida, India

To The New (TTN) 🔗

- Developed and deployed scalable backend services using **Java, Spring Boot, and Microservices Architecture** for a platform serving **11.8M+ users**.
- Implemented a pricing feature enabling **~56% of users** to upgrade channels, improving subscription workflows and user engagement.
- Designed and built a **promotions system** with multi-service integration, supporting configurable business rules and improving subscription conversion rates.
- Developed a **search and discovery service using Elasticsearch**, implementing multi-parameter queries, fuzzy search, filtering, and sorting to improve search relevance and performance.
- Built a **centralized data access microservice** integrating multiple downstream services, reducing redundant API calls and improving system efficiency.
- Managed **production deployments and workflows**, ensuring reliable releases across multiple microservices.
- Performed **production debugging and root cause analysis (RCA)** to resolve system issues and maintain high availability.

SKILLS

Programming Languages — Java (primary), Python (FastAPI)

Frameworks & Libraries — Spring Ecosystem (Spring Boot, Spring Security, Spring Data JPA, Hibernate)

Databases & Messaging — MySQL, PostgreSQL, Redis, Elasticsearch, RabbitMQ, Hazelcast

Cloud & DevOps — AWS (EC2, S3), Docker, Git, GitLab, Maven, Gradle

Testing & Observability — JUnit, Mockito, SLF4J Logging, Production Debugging

Core Concepts — Data Structures, OOP, SOLID, Microservices, Multithreading, REST API Design

PROJECTS

E-Commerce Backend System 🔗

Java, Spring Boot, MySQL, Spring Security, JWT

- Developed a **role-based backend system** (Admin, Seller, Customer) with secure authentication and authorization using **Spring Security and JWT**.
- Designed and implemented **RESTful APIs** for product management, user management, and catalog services.
- Modeled complex product structures with **metadata-driven design**, enabling flexible and scalable catalog management.
- Implemented **pagination, validation, and exception handling** for robust API design.
- Followed **layered architecture and clean code principles** for maintainability and scalability.

Doctor Appointment System 🔗

Python, FastAPI, MySQL, SQLAlchemy, JWT

- Built RESTful APIs for **appointment scheduling and booking systems** using FastAPI.
- Implemented **role-based access control (RBAC)** for doctors and patients with JWT authentication.
- Designed **time-slot scheduling system** with conflict detection and appointment lifecycle management.
- Used **ORM (SQLAlchemy)** for database interactions and modular architecture for maintainability.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

2021 – 2025

BML Munjal University

CGPA: 8.14 / 10